

Next-Generation Intelligence & Compliance

Empowering the IRS Office of Safeguards with a highly secure, intelligent, and accurate Retrieval-Augmented Generation (RAG) platform to navigate Publication 1075 and ever-evolving technical guidelines seamlessly.

The Challenge

The Office of Safeguards relies on **Publication 1075**, encompassing hundreds of pages. Traditional methods fall short:

- **Time-Intensive TI Research:** Producing a comprehensive Technical Inquiry response currently requires several hours of manual research per inquiry.
- **Interpretation Risk:** Reviewers must interpret complex guidance across Pub 1075, NIST standards, and technology-specific benchmarks, leading to inconsistent accuracy.
- **Search Limitations:** Standard keyword search misses semantic meaning and context necessary to parse complex agency questions.
- **Public AI Risks:** Commercial LLMs risk exposing **FTI** and **PII**, suffer from hallucinations, and cite outdated regulations.

The SkyShield Solution

We built a secure, private AI compliance platform engineered specifically for the IRS, bypassing commercial LLM vulnerabilities entirely.

- **Accelerated Research:** Significantly compresses TI research that typically takes hours down to minutes, dramatically increasing throughput.
- **Ironclad Security:** Intercepts and blocks FTI and PII *before* it ever reaches inference.
- **Zero Hallucination:** Strictly utilizes verified Pub 1075 and IRS data, enforcing factual, citation-backed responses.
- **Comprehensive Governance:** Built-in RBAC, user logging, and an executive dashboard for tracking and reporting.

Architecture & Methodology

Part 1: Data Ingestion & ETL

- **Dynamic Uploads:** Ingests Pub 1075, Interim Guidance, SCSEMs, and CIS benchmarks.
- **Agentic Contextual Chunking:** Uses a specialized micro-LLM to intelligently segment data, injecting context into each chunk to guarantee pristine retrieval.
- **Vector Database:** Stores high-fidelity embeddings in an encrypted Postgres vector database.



Part 2: Active AI Engine

- **Precision Retrieval:** Employs advanced semantic search to pull chunks, subsequently *re-ranking* them for definitive accuracy.
- **Robust Guardrails:** Full incident response logging actively shields all inferences against data leakage.
- **Review Workflows:** Enhances downstream Quality Control (QC) and Quality Review (QR) processes seamlessly.

Postgres Vector DB

Agentic Chunking

Re-Ranking

FTI/PII Guardrails

RBAC / Real-Time Auditing

Data Privacy

Transforming Technical Review Operations



EFFICIENCY

Reduces 1-2 hours of deep technical inquiry into minutes. Exponentially accelerates subsequent QC and QR processes.



ACCURACY

Relentlessly cross-references Pub 1075, NIST standards, and tech guides without missing critical dependencies.



CONSISTENCY

Replaces divergent individual reviewer opinions with a unified voice strictly backed by verifiable citations.



ANALYTICS

Comprehensive built-in dashboard provides executive visualizations, detailed usage logging, and real-time compliance tracking.